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STUDENTS RANKING SYSTEM

BY GROUP NUMBER 10.

System Description

PURPOSE

Practical application of sorting algorithms in a real-world scenario.

ALGORITHM

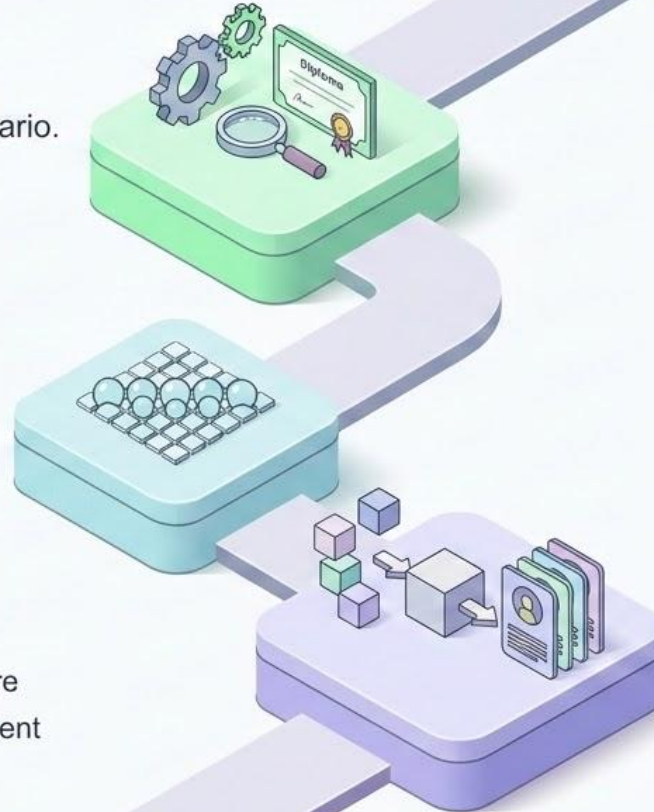
Bubble Sort

Efficiently re-orders student data based on updated performance metrics.

DATA FLOW

Input: Student name & new score

Output: Descending sorted student list by score



Key Features

- **Rank Visualization:** Clear display of original student rankings.
- **Dynamic Updates:** Modify student scores in real-time.
- **Automated Re-sorting:** Instant calculation and display of updated standings.



BUBBLE SORT

Visualization of Pass 1

Initial Array



Compare Adjacent Elements & Swap



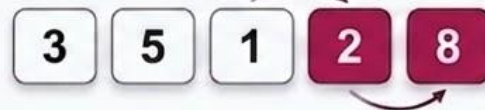
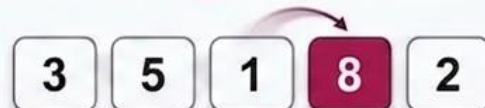
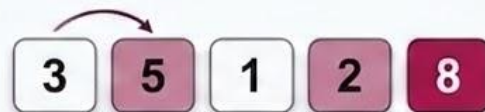
Compare. No Swap.



End of Pass 1: Largest number "bubbles" up and locks (8).

Visualization of Pass 2

1
2
3
4
5



Complexity

$$O(n^2)$$

```
for (int i = 0; i < n-1; i++) {  
    for (int j = 0; j < n-i-1; j++) {  
        if (arr[j] > arr[j+1]) {  
            // swap  
            int temp = arr[j];  
            arr[j] = arr[j+1];  
            arr[j+1] = temp;  
        }  
    }  
}
```

Repeat Until Sorted

Continue passes until no swaps are needed, indicating the entire array is sorted.





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Thank You